(iii) Average period method

If the partners withdraw fixed amount at fixed time interval, interest on drawings may be calculated on the basis of the average period. Fixed time interval refers to withdrawal made monthly, quarterly, half-yearly, once in 2 months and once in 4 months. The following formula may be used to calculate interest on drawings:

$$\text{Interest on drawings} = \left[\begin{array}{c}\text{Total amount of drawings}\\ \text{made during the year}\end{array}\right] \times \left[\begin{array}{c}\text{Rate of interest}\\ \text{per annum}\end{array}\right] \times \left[\frac{\text{Average period}}{12}\right]$$

$$\text{Average period} = \frac{\left[\begin{array}{c}\text{Number of months from}\\ \text{the date of first withdrawal}\\ \text{to the end of the year}\end{array}\right] + \left[\begin{array}{c}\text{Number of months from}\\ \text{the date of last withdrawal}\\ \text{to the end of the year}\end{array}\right]}{2}$$

Average period is computed as follows:

The following table shows the average period in months for withdrawal made at the beginning, in the middle and at the end of every month, quarter and half-year of the year.

Frequency of withdrawal	Average period (in months)		
	Time of withdrawal		
	At the beginning	In the middle	At the end
Monthly	$\frac{(12+1)}{2} = 6.5$	$\frac{(11.5+0.5)}{2} = 6$	$\frac{(11+0)}{2} = 5.5$
Quarterly	$\frac{(12+3)}{2} = 7.5$	$\frac{(10.5+1.5)}{2} = 6$	$\frac{(9+0)}{2} = 4.5$
Half-yearly	$\frac{(12+6)}{2} = 9$	$\frac{(9+3)}{2} = 6$	$\frac{(6+0)}{2} = 3$

Illustration 16

John is a partner in a firm. He withdraws ₹ 1,000 p.m. regularly. Interest on drawings is charged @ 5% p.a. Calculate the interest on drawings using average period, if he draws

(i) at the beginning of every month

(ii) in the middle of every month

(iii) at the end of every month

Solution

Total amount withdrawn = 1,000 × 12 = ₹ 12,000

(i) If drawings are made at the beginning of every month:

Average period = 6.5

$$\text{Interest on drawings} = \text{Total amount of drawings} \times \text{Rate of interest} \times \frac{\text{Average period}}{12}$$

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